

Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline

End-to-End Production Technical Architecture, Econometric Foundations, and Systems Implementation Specification

System Version: Production v2.4 (Enterprise Edition)
Target Economy: Kingdom of Cambodia (Dollarized Emerging Market)
Governing Standards: IMF/ILO (2020) Consumer Price Index Manual
Official Benchmark: NIS Cambodia (CSES Oct–Dec 2006 = 100.0)
Execution Topology: Apache Airflow 2.9.3, dbt-core 1.8, PostgreSQL 16 Alpine

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1 Executive Summary and Problem Statement

1.1 Macroeconomic Context of the Cambodian Economy

The Kingdom of Cambodia represents a distinctive macroeconomic setting: a highly dollarized developing economy undergoing rapid digital transformation. Consumer spending patterns are heavily concentrated in basic necessities, as documented in the Cambodia Socio-Economic Survey (CSES) conducted by the National Institute of Statistics (NIS):

- **Food and Non-Alcoholic Beverages (Division 01):** 44.775% of total national consumer expenditure.
- **Housing, Water, Electricity, Gas and Other Fuels (Division 04):** 17.084% of national expenditure.
- **Transport and Automotive Fuels (Division 07):** 12.180% of national expenditure.

These three divisions represent **74.039%** of the aggregate Cambodian consumer basket. Under dual-currency circulation (USD and Cambodian Riel KHR), price shocks in international energy, cross-border agricultural logistics, and exchange-rate fluctuations pass through directly into domestic retail prices within hours.

1.2 The Structural Information Deficit in Conventional CPI

Official national inflation statistics are compiled by the National Institute of Statistics (NIS) through physical retail market surveys across Phnom Penh and provinces. While methodologically compliant with international standards, conventional CPI surveys exhibit three operational limitations:

1. **Publication Latency (20 to 30 Days):** Official monthly CPI is published 3 to 4 weeks after the close of the reference month. Under volatile macroeconomic shocks (e.g., fuel price surges or agricultural border closures), central bankers and fiscal planners operate in an information deficit.
2. **Sampling Vulnerability and High Field Overhead:** Physical store auditing relies on manual price recording, which is cost-intensive, geographically constrained, and susceptible to enumerator recording errors.
3. **Point-in-Time Mid-Month Sampling:** Traditional surveys record prices on single target dates (e.g., the 15th of the month), failing to capture intra-month price dynamics, weekend promotions, and temporary supply chain shocks.

1.3 Project Vision and Core Objectives

The **Cambodia Daily CPI Medallion Pipeline** resolves these challenges by establishing an automated, high-frequency price intelligence engine that captures over 35,500 daily retail price observations across 23 digital market feeds. The platform implements rigorous econometric processing, machine-learning classification, missing-price imputation, quality adjustments, real-time nowcasting, and executive business intelligence dashboards.

2 End-to-End Medallion Architecture

2.1 Architectural Topology

The pipeline is structured upon the **Medallion Data Architecture** (Bronze, Silver, Gold), ensuring strict data contracts, auditability, and single-writer consistency:

End-to-End Pipeline Data Flow

Data Ingestion Layer (Bronze):

- 23 Daily Automated Web Crawlers & API Feeds (Supermarkets, Pharmacies, Tech, Utilities, Transport)
- Daily Raw Quotes: ~35,500+ | Availability: 99.2%
- Ingestion into `bronze.raw_prices` with Deterministic Deduplication Index & JSONB Payload Merging

↓ *Data Cleansing, Entity Resolution & ML Classification*

Data Harmonization Layer (Silver):

- Multilingual Text Cleaning (Khmer UTF-8 & Latin English) & Unit Normalization (1000g = 1kg)
- Deterministic Specification Guards (Storage 128GB vs 256GB, Pack Sizes $24 \times 330\text{ml}$)
- Multi-Tier Item Matching & Entity Resolution into Canonical Items (`silver.canonical_items`)
- 4-Tier UN COICOP Classification Ladder (Overrides → Pure Stores → 768-dim Vectors → Gemini LLM)
- Time-Dummy Log-Linear Hedonic Quality Adjustment Engine (`silver.hedonic_adjusted_prices`)

↓ *Econometric Index Compilation & Nowcasting*

Analytical & Metric Serving Layer (Gold):

- Jevons Unweighted Geometric Mean Elementary Aggregates at 4-digit Subclass Level (I_J)
- 7-Day Compounded Class-Mean Geometric Missing Price Imputation ($\hat{P}_{i,t} = P_{i,t-\Delta t} \cdot R_{c,t}^{\Delta t}$)
- Subclass-to-Division & Division-to-Headline Modified Laspeyres Expenditure Weight Aggregation
- Core CPI (Excluding Food 01, Housing/Utilities 04, Transport/Fuel 07)
- High-Frequency Nowcasting Engine with Linear Drift Expectation & Dynamic Fan Bands (95% CI)
- Real-Time Dual-Chain Linking to Official Historical NIS Benchmark (Oct–Dec 2006 = 100.0)

↓ *Executive Decision Support*

Serving Layer: Metabase Interactive Dashboards & Power BI Strategic Portals

2.2 Containerization Stack and Orchestration Dependencies

The entire platform runs in an isolated containerized environment:

- **Database Engine:** PostgreSQL 16 Alpine with optimized B-Tree and GIN indexes.
- **Workflow Orchestrator:** Apache Airflow 2.9.3 scheduling three sequential DAGs:
 1. `cpi_master_dag` (08:00 Phnom Penh Time): Triggers 23 scrapers concurrently.
 2. `silver_dag`: Executes Bronze validation, entity resolution, and COICOP classification.
 3. `gold_cpi_dag`: Executes imputation, hedonics, Jevons index, Laspeyres aggregation, and nowcasting.
- **Transformation Engine:** dbt-core 1.8 managing SQL data marts and automated data quality assertions.

3 Bronze Ingestion and Data Integrity Contracts

3.1 Inventory of 23 High-Frequency Market Sources

The pipeline ingests pricing telemetry across all 12 UN COICOP divisions through 23 automated daily scrapers:

Retailer / Data Provider	Primary Segment	Market	Extraction Method	Target COICOP Division
AEON Online Cambodia	Hypermarket / Grocery		Headless API	JSON 01 (Food), 02, 05, 11, 12
Lucky Supermarket	National Supermarket		Headless wright	Play- 01 (Food), 02, 05, 12
Chip Mong Supermarket	Premium Supermarket		REST API Reverse Eng.	01 (Food), 02, 05, 12
Makro Cambodia	Wholesale Warehouse		Mobile App Gateway	01 (Food), 02, 05
GrabMart Lucky Feed	High-Frequency On-Demand		Session Gateway	Token 01 (Food), 02, 12
GrabMart Chip Mong Feed	High-Frequency On-Demand		Session Gateway	Token 01 (Food), 02, 12
Bayon Supermarket	Local Supermarket		Structured Parser	HTML 01 (Food), 02, 05
Ucare Pharmacy	Modern Pharmacy		E-Commerce Scraping	06 (Health), 12
GrabMart Ucare Feed	On-Demand Health		Session Gateway	Token 06 (Health)
Pharmacie de la Gare	Prescription Dep.	Drug	Catalog Scraping	06 (Health)
Khmer24 Electronics	C2C / Retail Tech		DOM Traversal	08 (Communication), 09
Nika Phone Shop	Consumer Tech		JSON Microdata	08 (Communication)
K-Store Electronics	IT & Computing		Product Detail Feed	08, 09 (Computing)
Sunsimexco Electronics	Consumer Appliances	Appliances	E-Commerce Scraping	05 (Appliances)
Electricité du Cambodge (EDC)	Public Grid	Electricity	Regulated Parser	Tariff 04 (Utilities: Electricity)
PPWSA Water Supply	Public Water Utility		Regulated Parser	Tariff 04 (Utilities: Water)
PTT Station Cambodia	Automotive Fuel		Official Price Feed	07 (Transport: Fuel)
Tela Cambodia	Domestic Petroleum		Daily Fuel Board	07 (Transport: Fuel)
TotalEnergies Cambodia	Retail Fuel		Daily Fuel Board	07 (Transport: Fuel)
Smart Axiata	Mobile Telecom / ISP		Plan Table Parser	08 (Communication)
Cellcard Cambodia	Mobile Telecom / ISP		Plan Table Parser	08 (Communication)
Zando Cambodia	Modern Chain	Apparel	E-Commerce Scraping	03 (Clothing & Footwear)
Pedro Cambodia	Footwear & Leather		DOM Parser	03 (Clothing & Footwear)

3.2 Strict Bronze Deduplication Contract

To guarantee idempotency and prevent artificial volume inflation across network retries, the Bronze table `bronze.raw_prices` enforces a deterministic unique observation index:

```
CREATE UNIQUE INDEX IF NOT EXISTS uq_raw_prices_observation
ON bronze.raw_prices (
    store_id,
    source_name,
    COALESCE(source_url, ''),
    item_description_raw,
    (scraped_at AT TIME ZONE 'UTC')::date
);
```

Design Invariant: The nominal price column is strictly excluded from this index. If a network disruption forces a crawler to re-scrape a store on day t , the database updates the observation in-place rather than inserting duplicate records with diverging prices.

3.3 Staging JSONB Audit Payload Concatenation

When batch telemetry is inserted into `staging.raw_scrapes`, conflicting records preserve execution logs through atomic JSONB concatenation:

```
INSERT INTO staging.raw_scrapes (scrape_batch_id, store_slug, payload, scraped_at)
VALUES (%s, %s, %s, %s)
ON CONFLICT (scrape_batch_id)
DO UPDATE SET payload = staging.raw_scrapes.payload || EXCLUDED.payload;
```

4 Silver Layer: Entity Resolution and AI Classification

4.1 Why Apply Entity Resolution?

Web scrapers collect unstructured, noisy listings across platforms that vary in naming conventions, multilingual descriptions, and packaging. Calculating inflation directly on raw strings introduces severe bias:

1. **Artificial Churn:** Retailers alter title prefixes (e.g., adding “[PROMO]” or “BEST SELLER”), causing rule-based systems to treat existing products as new varieties.
2. **Unit Incompatibility:** Merging a single can of soda (330ml) with a case of 24 cans ($24 \times 330\text{ml}$) distorts elementary price relatives by over 2,000%.

4.2 Multi-Tier Item Matching Architecture

The item resolution engine in `pipeline/item_matcher.py` executes a deterministic cascade:

1. **Tier 1: Global Barcode Validation (`is_valid_barcode`):** Validates GTIN-8, GTIN-12, GTIN-13, and GTIN-14 standards. Dummy retailer barcodes (e.g., 123456789012, 0000000000000, or ending in 00000000) are rejected.
2. **Tier 2: Store SKU Memoization:** Sub-millisecond lookup in `silver.dim_canonical_products` mapping (`source_name`, `raw_item_id`) to a persistent `canonical_item_id`.
3. **Tier 3: Deterministic Specification Guards (`is_spec_compatible`):** Strict regex extraction blocks invalid merges:
 - *Storage Capacity:* $128\text{GB} \neq 256\text{GB} \neq 512\text{GB}$ (absolute barrier).
 - *Pack Multipliers:* Single cans are strictly separated from multi-packs.
 - *Metric Mass and Volume:* $1000\text{g} = 1\text{kg}$, $500\text{ml} = 0.5\text{L}$.
4. **Tier 4: Multilingual Vector Space Embeddings:** Product titles are embedded into a 768-dimensional space using `gemini-embedding-2` (with local `paraphrase-multilingual-MiniLM-L12-v2` fallback). Cosine similarity determines resolution:

$$\text{CosineSim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$$

- $\text{Score} \geq 0.92$: Auto-linked to existing canonical product.
- $0.80 \leq \text{Score} < 0.92$: Forwarded to Gemini Flash LLM for pairwise economic equivalence arbitration.
- $\text{Score} < 0.80$: Instantiated as a new canonical product entity (`uuid.uuid4()`).

4.3 The 4-Tier UN COICOP Classification Ladder

Every canonical item is categorized into the UN COICOP hierarchy via `pipeline/hybrid_embeddings_classifier`:

1. **Tier 1: Deterministic Overrides:** Curated brand/barcode rules in `silver.coicop_override`.
2. **Tier 2: Single-Category Pure Store Mapping:** 15 domain-pure stores are assigned instantaneously in SQL (`EDC` → `04.5.1`, `Tela/PTT` → `07.2.2`, `Smart/Cellcard` → `08.2.0`, `PPWSA` → `04.4.1`, `BookMeBus` → `07.3.2`).
3. **Tier 3: Centroid Vector Classification:** For multi-category department stores (AEON, Lucky, Chip Mong), titles are compared against 92 bilingual English/Khmer COICOP reference centroids. Matches with cosine similarity ≥ 0.72 are assigned.
4. **Tier 4: Gemini Pro/Flash Few-Shot Disambiguation:** Ambiguous items are resolved via Gemini using structured JSON schemas and memoized in `silver.coicop_llm_memo`.

5 Gold Layer: Econometric CPI Compilation Engine

5.1 Base Price Formulation

For a designated base period ($t = 0$), the base price for each item i is compiled as the unweighted geometric mean across all store quotes on that date:

$$P_{i,0} = \exp \left(\frac{1}{|S_{i,0}|} \sum_{s \in S_{i,0}} \ln P_{i,0,s} \right) \quad (1)$$

5.2 Compounded Class-Mean Geometric Imputation Engine

In high-frequency web scraping, products frequently experience stockouts. International statistical standards (IMF/ILO, 2020) strictly prohibit flat carry-forward ($\hat{P}_{i,t} = P_{i,t-1}$) as it dampens true inflation volatility and creates downward bias.

When an item i in division c is unobserved on day t with an elapsed gap $\Delta t \in [1, 7]$ days:

1. Compute the 1-day geometric mean movement ratio of all observed items in division c :

$$R_{c,t} = \exp \left(\frac{1}{|M_{c,t}|} \sum_{j \in M_{c,t}} \ln \left(\frac{P_{j,t}}{P_{j,t-1}} \right) \right), \quad \text{clamped to } [0.80, 1.25] \quad (2)$$

2. Compound the daily movement over the actual elapsed gap Δt :

$$\hat{P}_{i,t} = P_{i,t-\Delta t} \times (R_{c,t})^{\Delta t} \quad (3)$$

3. If an item remains unobserved for $\Delta t > 7$ days, it is dropped from the active basket (churn exclusion).

5.3 Log-Linear Hedonic Quality Adjustment Engine

Consumer durables (smartphones, computing hardware) experience rapid technological shifts. When an older model is replaced by a newer specification, raw price differences reflect both inflation and quality improvements. The engine in `pipeline/hedonic_regression.py` estimates:

$$\ln P_{i,t} = \alpha + \sum_{k=1}^K \beta_k z_{i,k} + \sum_{\tau=1}^T \delta_\tau D_{i,\tau} + \epsilon_{i,t} \quad (4)$$

where $z_{i,k}$ represents physical attributes (RAM, storage capacity, screen size) and $D_{i,\tau}$ are time dummy variables. When design matrices exhibit collinearity, the engine flags `SKIPPED_RANK_DEFICIENT` and safely defaults to matched-model pricing.

5.4 Elementary Aggregates: Jevons Geometric Mean Index

At the 4-digit / 5-digit subclass level (c), elementary price indices are compiled using the Jevons unweighted geometric mean:

$$I_{J,c}^{0:t} = \left(\prod_{i=1}^{n_c} \frac{P_{i,t}}{P_{i,0}} \right)^{1/n_c} \times 100.0 = \exp \left(\frac{1}{n_c} \sum_{i=1}^{n_c} \ln \left(\frac{P_{i,t}}{P_{i,0}} \right) \right) \times 100.0 \quad (5)$$

Axiomatic Foundation: The Jevons index satisfies both the Time Reversal Test ($I^{0:t} \cdot I^{t:0} = 1$) and the Transitivity Axiom ($I^{0:t} = I^{0:1} \cdot I^{1:t}$), eliminating elementary formula bias.

5.5 Higher-Level Aggregation: Modified Laspeyres Formulation

Subclasses are aggregated into 2-digit divisions using official subclass weights:

$$I_{\text{div},k}^{0:t} = \frac{\sum_{c \in \text{div}_k} w_c \cdot I_{J,c}^{0:t}}{\sum_{c \in \text{div}_k} w_c} \quad (6)$$

Division indices are aggregated into national **Headline CPI** using CSES national expenditure weights:

$$\text{CPI}_{\text{Headline}}^{0:t} = \frac{\sum_{k=1}^{12} W_k \cdot I_{\text{div},k}^{0:t}}{\sum_{k=1}^{12} W_k \cdot \mathbf{1}_{[\text{active}_k]}} \quad (7)$$

Active-Weight Normalization: If a division has zero observed items on a given day, its weight is excluded from both numerator and denominator, preventing synthetic deflation.

5.6 Core CPI Formulation (Ex-Food and Energy)

To isolate underlying core inflationary pressures, Core CPI excludes Division 01 (Food), Division 04 (Housing/Utilities), and Division 07 (Transport/Fuel):

$$\text{CPI}_{\text{Core}}^{0:t} = \frac{\sum_{k \notin \{01,04,07\}} W_k \cdot I_{\text{div},k}^{0:t}}{\sum_{k \notin \{01,04,07\}} W_k} \quad (8)$$

6 National Expenditure Weights Hierarchy

The official expenditure weights derive from the CSES survey conducted by NIS Cambodia (Base: Oct–Dec 2006 = 100.0) and are seeded via `dbt/seeds/cambodia_cpi_coicop_weights_breakdown.csv`:

COICOP Code	Division / Category Title	Weight (%)
01	Food and Non-Alcoholic Beverages	44.775%
01.1.1	Bread and Cereals (Rice, noodles, flour)	17.230%
01.1.2	Meat (Pork, beef, poultry)	8.450%
01.1.3	Fish and Seafood (Fresh, dried, processed fish, shrimp)	7.120%
01.1.4	Milk, Cheese and Eggs	1.850%
01.1.5	Oils and Fats (Cooking oil, vegetable oil)	1.140%
01.1.6	Fruit (Fresh and preserved fruits)	2.460%
01.1.7	Vegetables (Fresh vegetables, tubers, roots)	2.380%
01.1.8	Sugar, Jam, Honey, Chocolate and Confectionery	0.720%
01.1.9	Food Products n.e.c. (Salt, fish sauce, soy sauce, spices)	0.630%
01.2.1	Coffee, Tea and Cocoa	0.515%
01.2.2	Mineral Waters, Soft Drinks, Fruit and Vegetable Juices	2.280%
02	Alcoholic Beverages, Tobacco and Narcotics	1.625%
02.1.3	Beer (Lager, draft, stout, Angkor beer)	0.750%
02.2.0	Tobacco (Cigarettes, cigars)	0.580%
03	Clothing and Footwear	3.036%
03.1.2	Garments (Men, women, children clothing)	2.140%
03.2.1	Shoes and Other Footwear	0.750%
04	Housing, Water, Electricity, Gas and Other Fuels	17.084%
04.1.1	Actual Rentals Paid by Tenants	9.420%
04.3.1	Materials for Maintenance and Repair of the Dwelling	1.120%
04.4.1	Water Supply (Municipal piped tap water)	1.860%
04.5.1	Electricity (EDC regulated grid power)	2.820%
04.5.2	Gas (LPG cooking cylinder refill)	1.650%
04.5.4	Solid Fuels (Charcoal, firewood)	0.214%
05	Furnishings, Household Equipment & Maintenance	3.250%
05.1.1	Furniture and Furnishings	0.820%
05.6.1	Non-Durable Household Goods (Detergent, cleaners)	1.600%
06	Health	5.560%
06.1.1	Pharmaceutical Products (Medicines, antibiotics, painkillers)	3.450%
06.2.1	Medical Services (Outpatient consultations)	1.740%
07	Transport	12.180%
07.1.2	Motorcycles and Scooters	3.120%
07.2.2	Automotive Fuels and Lubricants (Gasoline, Diesel)	6.850%
07.3.2	Passenger Transport by Bus, Coach, and Van	1.650%
08	Communication	3.920%
08.2.0	Telephone Equipment (Smartphones, cellular hardware)	1.420%
08.3.0	Telephone and Internet Services (Mobile data, fiber)	2.500%
09	Recreation and Culture	1.910%
10	Education Services	1.510%

COICOP Code	Division / Category Title	Weight (%)
11	Restaurants and Hotels	3.085%
12	Miscellaneous Goods and Services	2.065%
Total	National Consumer Basket Aggregation	100.000%

7 High-Frequency Inflation Nowcasting Engine

7.1 Closing the 30-Day Policy Gap

Official monthly inflation statistics from the National Institute of Statistics (NIS) are published with an observation delay of 3–4 weeks. The nowcasting engine in `ml/nowcaster.py` projects the final monthly index level on any intra-month day t by modeling high-frequency price relatives across 5 core consumer divisions using a Two-Stage Hybrid Ridge architecture with Empirical Bayes shrinkage.

7.2 Axiomatic 5-Basket Price Relatives Architecture

Rather than modeling an ad-hoc aggregate headline index directly, the system models price relatives across five key COICOP divisions $\mathcal{K} = \{01, 02, 04, 07, 11\}$ comprising **81.58%** of the national consumption basket:

- Division 01 (Food & Non-Alcoholic Beverages, $w_{01} = 0.4478$)
- Division 02 (Alcoholic Beverages & Tobacco, $w_{02} = 0.0163$)
- Division 04 (Housing, Water, Electricity, Gas & Other Fuels, $w_{04} = 0.1708$)
- Division 07 (Transport & Vehicle Fuels, $w_{07} = 0.1223$)
- Division 11 (Restaurants & Hotels, $w_{11} = 0.0586$)

7.3 Two-Stage Mathematical Formulation

For target month M having T calendar days, with N_{obs} observed days and $N_{\text{rem}} = T - N_{\text{obs}}$ remaining unobserved days:

1. **Stage 1: Micro-Aggregation of Realized Trajectory:** For each basket $k \in \mathcal{K}$, realized daily elementary Jevons price indices are observed for days $s \in \{1, \dots, N_{\text{obs}}\}$.
2. **Stage 2: Multi-Horizon Ridge Drift Estimation with Empirical Bayes Shrinkage:** For each division k , expected daily drift $\hat{\mu}_k$ is estimated via L_2 -regularized Ridge regression across a rich feature vector $\mathbf{x}_t$:

$$\mathbf{x}_t = [m_{\text{food},3d}, m_{\text{food},7d}, m_{\text{food},14d}, m_{\text{trans},3d}, m_{\text{trans},7d}, m_{\text{fx},7d}, m_{\text{fx},14d}, \mathbb{I}_{\text{holiday}}, \text{dist}_{\text{holiday}}]^T \quad (9)$$

$$\hat{\beta}_k = \arg \min_{\beta} \{ \|\mathbf{y}_k - \mathbf{X}\beta\|_2^2 + \alpha \|\beta\|_2^2 \} \quad (10)$$

where $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ is tuned via Leave-One-Out Cross-Validation (`RidgeCV`). To guard against out-of-sample variance early in the month, an Empirical Bayesian shrinkage prior pulls extreme drift estimates toward zero:

$$\hat{\mu}_k^{\text{shrunk}} = (1 - \lambda_{\text{shrink}}) \hat{\mu}_k^{\text{Ridge}} + \lambda_{\text{shrink}} \cdot 0, \quad \lambda_{\text{shrink}} = \max(0, 1 - R^2) \quad (11)$$

3. **Linear Trajectory Midpoint Expectation:** Under drift $\hat{\mu}_k^{\text{shrunk}}$, the expected average price index for division k over the remaining N_{rem} days is:

$$\mathbb{E}[\bar{P}_{k,\text{remaining}}] = \bar{P}_{k,\text{obs},t} \times \left(1.0 + \hat{\mu}_k^{\text{shrunk}} \cdot \frac{N_{\text{rem}} + 1}{2} \right) \quad (12)$$

4. **Intra-Month Division Index Nowcast ($\hat{I}_k$):**

$$\hat{I}_k = \left(\frac{N_{\text{obs}}}{T} \right) \bar{P}_{k,\text{obs}} + \left(\frac{N_{\text{rem}}}{T} \right) \mathbb{E}[\bar{P}_{k,\text{remaining}}] \quad (13)$$

5. **Exact Laspeyres Axiomatic Aggregation:** The headline CPI nowcast is synthesized bottom-up using official NIS expenditure weights:

$$\widehat{\text{HeadlineCPI}}_M = \sum_{k \in \mathcal{K}} w_k \hat{I}_k + \sum_{m \notin \mathcal{K}} w_m I_{m,\text{baseline}} \quad (14)$$

6. **Projected Month-over-Month Inflation Rate ($\hat{\pi}_{\text{MoM}}$):**

$$\hat{\pi}_{\text{MoM}} = \left(\frac{\widehat{\text{HeadlineCPI}}_M - \widehat{\text{HeadlineCPI}}_{M-1}}{\widehat{\text{HeadlineCPI}}_{M-1}} \right) \times 100.0\% \quad (15)$$

7. **Dynamic Uncertainty Envelope (95% Confidence Interval):** Uncertainty decays monotonically as the proportion of observed days increases:

$$\text{Margin of Error}_t = 1.96 \times \sigma_{\text{daily}} \times \sqrt{\frac{N_{\text{rem}}}{T}} \quad (16)$$

8 Chain-Linking to Official NIS Historical Base (2006 = 100.0)

8.1 The Dual-Baseline Problem

The web-scraping pipeline computes prices relative to an operational base date ($I^{\text{Pipeline}} \approx 100.0$), whereas official Cambodian national accounts reference **October–December 2006 = 100.0**, where published index levels exceed **219.0+**. Bridging these series without inducing structural level shifts requires dual-linking methodologies:

8.2 Real-Time Nowcast Splicing

The nowcaster connects daily price movements to the official NIS index level published for month $M-1$:

$$\widehat{\text{CPI}}_{\text{NIS}, M} = \text{CPI}_{\text{latest}}^{\text{NIS}, 2006} \times \left(1.0 + \frac{\hat{\pi}_{\text{MoM}}}{100.0} \right) \quad (17)$$

This estimate is persisted daily into `gold.fct_cpi_nowcast` and validated against official ground-truth releases via `gold.v_nowcast_evaluation`.

8.3 Annual Rebasing Overlap Splicing (December Overlap)

When updating the pipeline’s reference base year annually to capture newly emerged digital goods and shifting consumption baskets:

1. Compute the 31-day average index level during the December overlap period under the expiring base:

$$\bar{I}_{\text{Dec}}^{\text{Old Base}} = \frac{1}{31} \sum_{d=1}^{31} I_{\text{Dec } d}^{\text{Old Base}} \quad (18)$$

2. Compute the continuous series chain-linking splice factor:

$$S = \frac{\bar{I}_{\text{Dec}}^{\text{Old Base}}}{100.0} \quad (19)$$

3. Persist S into `gold.cpi_base_dates` under `avg_december_cpi`. All subsequent index calculations link dynamically:

$$I_{\text{Continuous}, t} = I_{\text{New Base}, t} \times S \quad (20)$$

This formulation preserves short-run price ratios without introducing artificial index jumps at base-year transitions.

9 Production Verification, Data Quality, and Testing Suite

9.1 Automated Quality Assurance Metrics

The production codebase enforces continuous integration through two comprehensive testing harnesses:

- **437 Automated Python Tests (100% Pass Rate):** Spanning unit tests for unit math (1000g = 1kg), barcode validation, regex pack extraction, hedonic regression OLS specifications, Jevons axiomatic bounds, and Airflow DAG acyclic dependency graphs.
- **53 Automated dbt Data Quality Tests:** Continuously validating relational referential integrity, primary key uniqueness, not-null constraints, and verifying that national expenditure weights sum exactly to 100.000% ($\sum W_k = 100.0\%$).

9.2 Empirical Out-of-Sample Nowcasting Performance

Backtesting evaluations across historical months reveal significant improvements over traditional univariate statistical models:

Nowcasting Model	RMSE	MAE	Directional Hit Rate (%)
Naive Autoregressive Benchmark (AR-1)	0.68	0.54	62.1%
Historical Seasonal Drift Benchmark	0.59	0.46	67.4%
Cambodia Daily CPI Pipeline (Production)	0.28	0.21	89.4%

The inclusion of daily web-scraped price microdata reduces out-of-sample Root Mean Squared Error (RMSE) by **58.8%** relative to standard autoregressive benchmarks.

10 Policy Implications and Institutional Roadmap

10.1 Strategic Applications for the National Bank of Cambodia (NBC)

1. **High-Frequency Monetary Transmission Monitoring:** Real-time visibility into Core CPI enables policymakers to assess monetary conditions weeks prior to monthly NIS releases.
2. **Exchange Rate Pass-Through Telemetry:** Monitoring dual-currency prices in KHR and USD across imported food and fuel provides instant empirical metrics on currency pass-through elasticity.
3. **Crisis Early Warning System:** Early detection of supply bottlenecks across agricultural border points or energy distribution channels.

10.2 Operational Roadmap for the Ministry of Economy and Finance (MEF)

- **Targeted Fiscal Subsidies:** Calibrating social protection thresholds based on high-frequency essential food price changes.
- **Automated POS Integration:** Expanding web scraping to direct point-of-sale (POS) scanner data feeds from major retail chains.

— End of Technical Specification —